Security Risk Assessment: Reconstructing a Key from a Low-Quality Photograph with Help from AI

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Abstract

Mechanical keys can expose sensitive access information without being lost or stolen: a photograph may reveal enough of a key's geometry to support its reconstruction. This article examines the security risks associated with reconstructing a key from a low-quality photograph and considers how artificial intelligence can lower the technical barriers involved. It describes the use of reference specifications and visual decoding tools, while addressing the uncertainty introduced by incomplete or unclear images. Particular attention is given to master-key systems, where the compromise of a single key can affect access to numerous spaces. Beyond the technical vulnerability, the discussion examines key exposure during everyday work, the assignment of access privileges, and the responsibilities placed on employees who carry highly privileged keys. It argues that effective protection requires more than individual vigilance: practical handling procedures, appropriate training, and organizational support must accompany the access granted. AI is presented not as the source of a new vulnerability, but as a tool that can make an existing one more accessible, reinforcing the need to reassess both technical safeguards and key-management practices.

1 Introduction

One of the first methods I learned for copying a key from a photograph uses a drawing program. The photograph is placed on the canvas, and a key decoder is positioned over the image. A key decoder is a visual reference made of lines placed at specific intervals. Each line corresponds to a cut depth in a key system. When the decoder is correctly aligned with the key in the photograph, the position of each cut can be compared with the reference to estimate its depth.

For common key systems, key decoders can sometimes be found online. Examples include standard types such as American, Arrow, BEST, Kwikset, Sargent, Schlage, and Yale. However, a decoder may be difficult to find for a less common or more specific key type, such as the Old Russwin 752. Because there are many different key systems, people may have little reason to create and publish a decoder for every one of them. This can make finding a suitable decoder difficult.

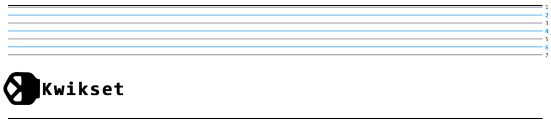


Figure 1: Key Decoder for Kwikset. Work of Deviant Ollam. [5]

Creating a decoder for an unfamiliar key system is also challenging. It requires accurate proportions, reliable reference information, and a good understanding of the key system and its master-key system. With the arrival of artificial intelligence, however, a beginner can ask an AI system to help create a decoder for a specific key type. As a result, the beginner may no longer need to search through large amounts of technical documentation to find the information needed to interpret a key from a photograph.
In this article, we will examine methods people use to copy a key from an image, with the goal of identifying the weaknesses they may exploit and considering how key systems could be improved to reduce this risk.
But first, if you are unfamiliar with master key systems, you may want to read Matt Blaze's paper [3], which explains them very clearly at the pinning level.

2 Obtaining a Photograph of a Key

Before examining how key information may be inferred from a photograph, three questions must be considered:

1. Which key would present the greatest security risk if photographed?
2. Where might that key be found, and under what circumstances might it be exposed?
3. What characteristics make a photograph easier to analyze?

2.1 Choosing a Key

If the objective is to understand the greatest potential exposure from a single key, a **master key** or, where one exists, a **grand master key** would be of particular concern. A master key generally operates a group of locks, while a grand master key may operate several groups within a larger system. The exact meaning of these terms depends on the hierarchy designed for the facility.
If the concern is limited to a particular room or area, the relevant key may instead be the key assigned to that lock. In practice, however, the risk is not limited to high-level keys: the method of analyzing an image may apply to many types of keys, provided that enough information about the key system is available.

2.2 Where Key Exposure Can Occur

To understand how a master key might become exposed to unauthorized photography, one must first consider who holds such a key and how it is handled during everyday work. Common keyholders include custodial staff, maintenance personnel, managers, and security officers.

From an adversarial perspective, employees with predictable routines may present particular opportunities. Their movements are easier to anticipate, and repeated exposure to familiar environments can make security precautions feel less urgent. Custodial and security personnel are therefore relevant to this discussion, although their responsibilities, working conditions, and security training may differ considerably.

Custodial staff deserve particular attention, not because they are inherently less trustworthy, but because their work can combine extensive physical access with demanding tasks and limited security-specific training. Their primary responsibility is to clean and maintain facilities, rather than to monitor suspicious behavior. Concentration on these duties, sometimes while wearing headphones, may reduce their awareness of nearby activity. Under such conditions, an unattended key can become an opportunity for unauthorized photography.

Custodial work illustrates the tension between security procedures and operational demands. Employees may be expected to follow the same precautions hundreds of times during a working day: removing keys from locks immediately after use, keeping them securely attached to their person, and never leaving them unattended on a cleaning cart.

Although individually straightforward, these actions create a recurring burden when performed throughout a shift. If their purpose is not clearly explained, or if compliance interferes with the practical demands of the job, employees may gradually develop shortcuts. Keys may occasionally remain in a door, be placed on a cart, or otherwise be left unsecured while the employee focuses on another task.

Such situations can create opportunities for someone nearby to photograph a key without authorization. The underlying issue, however, should not be reduced to individual carelessness. A security arrangement that depends on uninterrupted vigilance throughout physically demanding and repetitive work must also be examined as an organizational problem.

Custodial personnel perform essential and demanding work. When an organization assigns them highly privileged keys, it also assigns them a substantial security responsibility. That responsibility must be supported by appropriate training, realistic procedures, and suitable arrangements for carrying and storing keys. For further details, see Section 4.1 and 4.1.

2.3 Photograph Characteristics

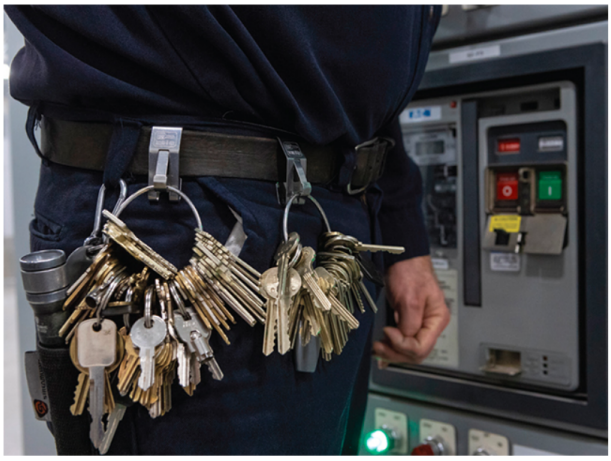
A photograph is easier to analyze when the key is clearly visible and its cuts can be distinguished. A key lying flat on a surface is generally easier to inspect than one photographed at an angle. A gridded background can also provide a visual reference for judging proportions.

However, an ideal photograph is not necessarily required for an image to reveal useful

information. Depending on the key type and the image, even a poor photograph may expose details that should be treated as sensitive. This is why key-control procedures should account for incidental images, not only photographs deliberately taken for duplication.

2.4 Sometime you dont need to take any photograph, they will do it for you!

It is not always necessary for an unauthorized person to take a photograph of a key: someone else may already have published one. Employees may inadvertently expose their keys in photographs shared on social media. For example, a security officer proud of a new position might post a photograph of their keys and identification badge on Facebook to announce their new job, without recognizing the potential security implications.
A similar situation can arise in journalism. When preparing a profile of an employee, a journalist might photograph the keys hanging from the employee's belt as a visual representation of their responsibilities. Although this may seem like an effective photographic choice, it can unintentionally reveal security-sensitive details.
One example appears in an article published by *Affaires universitaires*, entitled *Incursion dans le quotidien du service d'entretien d'une université*, which includes a photograph of an employee's key ring.[4] It illustrates how photographs intended simply to document everyday work can also expose details of physical access arrangements.



Un gros plan des trousseaux de clés portés par Réal Tardif.

Figure 2: Photograph of an Ulaval employee's key ring fond on a article published by *Affaires universitaires*.[4]

3 Analyzing Key Photographs

Analyzing a photograph of a key requires a decoding chart, suitable drawing software, and some hands-on work. The first step is to prepare a reference chart for the relevant key system.

3.1 Preparing a Decoding Chart

A decoding chart is based on the manufacturer's cutting specifications, particularly the table of nominal cut depths, sometimes referred to as a *bitting chart*. This table provides the reference measurements associated with the different depth codes used in a particular key system. Before selecting these specifications, the key system must be identified. Markings on the key may help with this identification, although they should not be treated as conclusive on their own. Once the appropriate specifications have been obtained, they can be used to prepare a decoding chart for an authorized analysis. An AI assistant may help format the reference data or draft a script. Alternatively, the following Python script generates a transparent PNG reference overlay from measurements supplied in a JSON configuration file. The configuration specifies the image scale, the cut positions relative to the key shoulder, and the reference heights measured from the back of the key. It also allows the margins, line color, and line thickness to be adjusted.
The script and its configuration file are available in the accompanying GitHub repository: Key-Decoder.[6] All files in this repository were created using AI, demonstrating how easy it is nowadays to build your own key decoder with AI assistance.

3.2 Analyzing the Photograph

The next step is to open the photograph in an image-editing application that supports layers and transformation tools. The free version of Sketchbook is used here, although other applications with equivalent features may also be suitable.
Place the photograph of the key on the first layer and enlarge it to obtain a clear working view. Import the decoding overlay onto a second layer, above the photograph.
Next, use the transformation tool to position and align the overlay with the key image. Once the overlay is aligned, read the reference depth corresponding to each cut.

3.3 Cutting the Key Blank

Once the bitting has been determined, the final step is to obtain a compatible key blank and cut it according to the identified depths. This can also be done by manually filing the blank. The result is a physical copy of the key, provided that the measurements, blank selection, and final cuts are accurate.
If you are unsure of the depth of one or even two cuts, you can use some of the techniques described here. The following examples illustrate how to handle different cases knowing that even if you cannot determine the exact depth from the photograph, you may be able to estimate it to within one depth increment.

3.3.1 Uncertainty about a single cut depth

If you are unsure of the cut depth at only one pin position:

1. Cut all the other positions to their corresponding master-key depths, leaving the uncertain position unfinished.

2. Gradually deepen the cut at the uncertain position, testing the key after each increment. Start slightly shallower than the depth you estimate from the photograph: you can always remove more material, but you cannot put it back.

3. When the lock opens, you have identified a working cut depth at that position. If this is the master-key depth, you now have a working master key.

3.3.2 Uncertainty about two or more cut depths

If you are unsure of the depths at two different pin positions, the method proposed here requires a change key, that is, a user key for the lock.

1. Cut all the known positions to their corresponding master-key depths, leaving the two uncertain positions unfinished.

2. At one of the two uncertain positions, cut to the depth found on the change key.

3. Gradually deepen the cut at the other uncertain position, testing the key after each increment. As before, start slightly shallower than the depth estimated from the photograph. This step allows you to identify a working cut depth at that position.

4. To determine the master-key depth at the position used in Step 2, repeat the incremental cutting procedure at that position on a new key blank, using the depths already determined for the other positions.

3.3.3 No information about one pin position

If one cut is completely obscured in the photograph and you have no information about its depth:

1. Cut all the known positions to the depths determined using the key decoder, leaving the unknown position unfinished.

2. Gradually file down the cut at the unknown position, testing the key in the lock after each depth increment until the lock opens. Remember that material can be removed, but not replaced.

3. At this point, you have found a working depth, but you do not yet know whether it corresponds to a change-key depth or the master-key depth. Assuming that all the other cuts are already at their master-key depths, test the key in another lock operated by the same master key. If it opens that lock as well, this supports the conclusion that you have found the master-key depth. If it does not, return to Step 2 and continue the process.

3.3.4 No information about multiple pin positions

If you have no information about the cut depths at more than one pin position, the most practical option may simply be to obtain another photograph. As discussed earlier, opportunities to photograph the master key may not be difficult to find. Also, you can use some of the techniques described by Matt Blaze [3].

4 Discussion

The compromise of a master key or grand master key (if there's one) can have serious consequences. If such a key falls into unauthorized hands, the impact extends beyond physical security: restoring the integrity of the affected keying system may also involve substantial financial costs.
The scope of the compromise depends on the access privileges associated with the key. In the case of a grand master key, a single incident may affect a large portion of a building's keying system, potentially requiring extensive rekeying or replacement of affected lock cylinders.
How can these risks be reduced? The first step is to examine the vulnerabilities that an unauthorized person could exploit. This chapter then considers potential safeguards, with some attention to reducing opportunities for unauthorized photography of keys with extensive access privileges. Ultimately, I do not intend to offer specific recommendations. Although I have gained considerable experience in this area, my conclusions may still prove mistaken, and I do not consider it my responsibility to provide professional security advice free of charge. More importantly, security requirements depend heavily on the environment in which a system is used. Measures that are appropriate in one setting may be unsuitable in another, so recommendations should not be assumed to apply universally.

4.1 Assigning Access Privileges

A grand master key (GMK) may be issued to custodial staff because they need to enter many rooms. From an operational standpoint, a single key that opens all relevant doors appears convenient: it reduces the number of keys carried and simplifies movement through the building. However, that convenience must be weighed against the consequences of the key being lost, photographed, or otherwise compromised.
The need to open many doors should not, by itself, justify possession of the most privileged key in a system. Access should instead reflect the employee's actual duties, the areas they must enter, the accountability associated with the role, and the security measures available to support that access.
Where broad access is genuinely necessary, issuing a highly privileged key should be accompanied by clear handling procedures and appropriate training. Where access requirements are narrower, several keys with more limited privileges may be preferable to a single grand master key. The objective is to provide sufficient access for the work without unnecessarily increasing the consequences of a single compromise.
you really need a grand master key? The central concern is not merely whether an unauthorized person can obtain or photograph a key, but how much access that key

could provide. A lapse involving a limited-access key and a lapse involving a grand master key can have very different consequences.
Depending on the design of the keying system and the extent of the compromise, an exposed grand master key could require extensive rekeying or replacement of affected cylinders. Assigning access more narrowly can help limit that exposure.
The broader lesson is that key security cannot rely solely on asking employees to remain vigilant. It must also account for how people actually work. Appropriate access privileges, practical handling procedures, and adequate training should reinforce one another, rather than leave individual workers responsible for compensating for weaknesses in the overall system.

4.2 Where Does AI Fit into All This?

Developing a key decoder with limited technical knowledge can take a considerable amount of time. Uncertainty about whether it will even work may discourage people from attempting this approach in the first place. With AI assistance, however, creating a key decoder is easier than ever.
This also raises a broader question: how might AI have changed other methods of bypassing key-based security systems?

4.3 Your Locksmith Is Not a Physical Security Consultant

Here is a piece of advice I like to share whenever I have the opportunity: your locksmith is not a physical security consultant.
If you look at the courses required for a vocational diploma in locksmithing, none really focuses on security itself. [8] Instead, they cover the installation of different types of locks and security systems, customer relations, and health and safety. Locksmiths are not expected to know every vulnerability in the systems they install, and that is understandable. Their job is to install those systems and prepare the charts needed to design coherent master key systems. The workers building a bridge do not need to understand all the structural calculations performed by the civil engineer in order to build it.
Assessing those vulnerabilities is the role of a physical security consultant. Yet there is no clearly defined educational path into that profession. The field is something of a jungle: it is unregulated, and anyone can call themselves a "security expert" without having to demonstrate the expertise that title implies.
Becoming a physical security consultant means proving your abilities by uncovering vulnerabilities, gaining hands-on experience, continually researching new ways to bypass security systems, and developing an in-depth understanding of how they work.
The rest of this section is more of an opinion piece. If you would like to read it, you can find it on my personal website: https://gabrielmarie.ca/Lock/article/locksmith-not-security-consultant.html[7].

4.4 Some More Concrete Recommendations

To help protect against this type of vulnerability, one could imagine a simple piece of fabric attached to a keyring that drapes over the keys, concealing them while they hang from the user's belt loop.
Another possibility would be a retractable metal sleeve fitted to the key, covering it when not in use and retracting naturally as the key is inserted into the lock. This would not be particularly difficult to implement if only a small number of master keys were in circulation within the building.

Reference

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